AI FITTRACK API

AI-Powered Fitness Tracking and Recommendation System

PROJECT PRESENTATION

20-Slide Presentation

INTRODUCTION

- AI FitTrack API is a smart fitness tracking system.
- It collects fitness and activity information through APIs.
- AI techniques can analyze user data and provide useful recommendations.
- The system is designed to make fitness monitoring simple and organized.

PROBLEM STATEMENT

- Manual fitness tracking can be time-consuming.
- Users may find it difficult to monitor different activities in one place.
- Basic tracking systems may provide limited personalization.
- A smart API-based system can automate data collection and analysis.

OBJECTIVES

- Track daily fitness activities and workout information.
- Store and manage user fitness data.
- Provide personalized fitness suggestions.
- Use APIs for communication between application components.
- Improve fitness monitoring and user experience.

EXISTING SYSTEM

- Fitness information is often recorded manually.
- Different applications may store different activity details.
- Limited systems provide personalized recommendations.
- Data analysis may require additional manual effort.

PROPOSED SYSTEM

- AI FitTrack API provides a centralized fitness-tracking service.
- Users can record workouts, activities and basic fitness details.
- API endpoints allow applications to send and receive data.
- AI-based analysis can support personalized recommendations.
- The system can be extended with mobile or web interfaces.

KEY FEATURES

- User profile and fitness data management
- Workout and activity tracking
- Fitness history and progress monitoring
- AI-assisted recommendation
- REST API communication
- Secure data handling

SYSTEM ARCHITECTURE

- User / Mobile or Web App

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- AI FitTrack REST API

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- Authentication → Business Logic → AI Analysis

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- Fitness Database

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- Personalized Results

MODULES

- User Management – handles profile and basic fitness information.
- Activity Tracking – records daily activities and workouts.
- Workout Management – stores exercise details.
- AI Recommendation – analyzes data and generates suggestions.
- API Management – provides endpoints for application communication.
- Database – stores fitness records and history.

API OVERVIEW

- API stands for Application Programming Interface.
- AI FitTrack API uses request and response communication.
- Client applications send HTTP requests to endpoints.
- The server validates, processes and stores data.
- The API returns structured responses, commonly in JSON format.

SAMPLE API ENDPOINTS

- POST /api/users – create a user
- GET /api/users/{id} – get user information
- POST /api/workouts – add workout details
- GET /api/workouts/{id} – view workout history
- POST /api/recommendations – request fitness recommendation

DATABASE DESIGN

- Users: user_id, name, age, activity_level
- Workouts: workout_id, user_id, exercise, duration, calories
- Activities: activity_id, user_id, steps, distance, date
- Recommendations: recommendation_id, user_id, message
- Relationships connect user records with fitness activities.

AI COMPONENT

- AI can analyze activity and workout patterns.
- Input data may include workout duration, activity level and history.
- The analysis can identify patterns and provide suitable suggestions.
- Recommendations should be treated as general fitness guidance, not medical advice.
- The AI component can be improved using trained machine-learning models.

WORKING PROCESS

1. User enters or syncs fitness information.
2. Client sends data to the API.
3. API validates the request.
4. Data is stored in the database.
5. AI module analyzes relevant information.
6. Recommendation is returned to the user.

TECHNOLOGIES USED

- Frontend: HTML, CSS, JavaScript / mobile UI
- Backend: Python with Flask or FastAPI
- API: REST and JSON
- Database: MySQL / SQLite
- AI/ML: Python and suitable machine-learning libraries
- Development: Visual Studio Code

SECURITY & VALIDATION

- Validate user input before processing.
- Use authentication for protected endpoints.
- Apply secure password storage.
- Limit access to authorized user data.
- Handle invalid requests with clear API responses.
- Avoid exposing sensitive information in responses.

TESTING & SAMPLE OUTPUT

- Test user creation and profile retrieval.
- Test workout creation and history retrieval.
- Test invalid API requests.
- Test recommendation generation.

Sample response: { "status": "success", "message": "Workout saved" }

ADVANTAGES & FUTURE SCOPE

- Centralized fitness data management
- Easy integration with web and mobile applications
- Personalized fitness recommendations
- Scalable API-based architecture
- Future: wearable-device integration
- Future: advanced AI analytics and progress dashboards

CONCLUSION & REFERENCES

• AI FitTrack API provides a structured approach to fitness tracking.

• It combines REST API communication, database management and AI-based analysis.

• The project can be expanded into a complete fitness application.

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• References: Python Documentation, Flask/FastAPI Documentation, REST API Documentation, Scikit-learn Documentation